CS5002
SP 2026
TEMPLATE

Problem 1. (**5 points**) Write out Row 11 of Pascal's triangle.

Answer: 1, 10, 45, 120, 210, 252, 210, 120, 45, 10, 1

Problem 2. (**10 points**) Compute the following:

(a) 5 choose 3 $\rightarrow$ **10**

(b) $\binom{10}{2} \rightarrow$ **45**

(c) 7 P 4 $\rightarrow$ **840**

(d) 1234 P 0 $\rightarrow$ **1**

(e) Number of ways you can re-arrange the letters in the word "JUPITER" $\rightarrow$ **7! = 5040** *Note, this last one is a permutation, not a combination (there'd only be 1 for a combination anyway)*

Problem 3. (**10 points**) This problem involves 8-digit binary strings such as 10011011_2 or 00001010_2 (8-digit numbers composed of 0's and 1's).

(a) How many such strings have 1's for their second **and** fourth digits?
Answer: 2 * 2 * 2 * 2 * 1 * 2 * 1 * 2 = **64**

(b) How many such strings have 1's for their second **or** fourth digits?

- You can do this in 2 different ways:
- Subdivide into 3 ways:
  - A = binary strings where the second digit is a 1 (0,1, 0,1, 0,1, 0,1, 0,1, 0,1, **1**, 0,1)
  - B = binary strings where the fourth digit is a 1 (0,1, 0,1, 0,1, 0,1, **1**, 0,1, 0,1, 0,1)
  - C = binary strings where BOTH the second and fourth digits are 1: 0,1, 0,1, 0,1, 0,1, **1**, 0,1, **1**, 0,1)
  - $|A|$ + $|B|$ - $|A \cap B|$ = 2^7 + 2^7 - 2^6 = **192**
- or subdivide in 2 ways:
  - A = binary strings where the second digit is a 1 (0,1, 0,1, 0,1, 0,1, **0**, 0,1, **1**, 0,1) (|A| = 2^6)
  - B = binary strings where the fourth digit is a 1 (0,1, 0,1, 0,1, 0,1, **1**, 0,1, **0**, 0,1) (|B| = 2^6)
  - C = binary strings where BOTH the second and fourth digits are 1: 0,1, 0,1, 0,1, 0,1, **1**, 0,1, **1**, 0,1) (|C| = 2^6)
  - In this case, you can just do |A| * |B| * |C| which is still 192

Problem 4. (**12 points**) For this problem, consider a 3-digit combination lock with the numbers 0-9

(a) There are many codes for which the three digits are all odd numbers, including 173, 911, and 555. Determine the total number of codes for which the three digits are all odd numbers. Explain your answer.

- **Answer:** There are 5 odd numbers: 1,3,5,7,9. For 3-digit code where the number can repeat: because every digit will have 5 possibilities, the total number would be: 5^3 = **125**

(b) There are many codes for which the three digits are all different, including 801, 097, and 123. Determine the total number of codes for which the three digits are all different. Explain your answer.

- **Answer:** For 3-digit code where the number cannot repeat: Because the number cannot be the same for each digit, so for the first digit, here are 10 (0-9) possibilities, and the second is 9 (subtract the one which was used in the first digit) and therefore the third is 8.
- The total: 10 * 9 * 8 = **720**

(c) There are many codes for which the three digits are all different and in increasing order, including 123, 059, and 479. Determine the total number of codes for which the three digits are all different and in increasing order.

- **Answer:** The first number has to be between 0 and 7 to suit our criteria (8 only has 1 single digit number larger and 9 has zero)
- For a code that starts with 0 there are 36 solutions
  - 0 1 {2/3/4/5/6/7/8/9} $\rightarrow$ 8 choices for 3rd digit
  - 0 2 {3/4/5/6/7/8/9} $\rightarrow$ 7 choices for 3rd digit
  - all the way up to 0 8 9 $\rightarrow$ 1 choices for 3rd digit
  - so we do 8 + 7+ 6 + 5 + 4 + 3 + 2 + 1 = 36
  - put another way, that's $\sum_{i=1}^{8} i$
- Codes that start with 1 have 8 fewer choices (since 0 can't be a subsequent number so we do $\sum_{i=1}^{7} i$
- So we end up 36 + (36 - 8) + (36 - 8 - 7) + ... and so on until that last digit is 1
- 36 +28 +21 + 15 + 10 + 6 + 3 + 1 = 120
- **Alternate answer:** Weirdly, any combination without repetition (if arranged in ascending order) $\binom{10}{3}$ also ends up being fine. A bit odd as an approach but a still legal explanation.

Problem 5. (**20 points**) I have a box with 50 candy bars. 8 kids come over on Halloween night, to trick-or-treat.

(a) How many ways can I distribute all 50 candy bars to the 8 kids?

- 50 candy bars, 8 children $\rightarrow$ 7 partitions
- $\binom{57}{50}$ or $\binom{57}{7}$, same result
- **Answer:** 264,385,836 (or just $\binom{57}{50}$ is fine)

(b) How many ways can I distribute all 50 Snickers candy bars to the 8 kids if every kid gets at least 2 candy bars?

- After each kiddo gets 2, there are 34 left over. We still need 7 partitions.
- **Answer:** $\binom{34+7}{34} = \binom{41}{34}$ = 22,481,940

Problem 6. (**11 points**) You shuffle a standard deck of 52 playing cards and then draw 15 cards. What is the largest number of cards that I can guarantee will share a suit?

- **Answer: 4**
- By the pigeonhole principle: If n objects are placed in k boxes, then there must exist a box that contains $\lceil n/k \rceil$ or more objects
- in a best-case scenario, I'd draw 13 clubs and 2 of whatever else, but I can't guarantee that sort of luck
- in a *worst* case scenario, I'd have the most even distribution of cards across all 4 suits,
- I have 15 cards (pigeons) and 4 suit "buckets" to put them in $\lceil 15/4 \rceil$ = 4
- In an even distribution, (Ex. 4 clubs, 4 hearts, 4 diamonds, and 3 hearts)

Problem 7. (**12 points**) A salesman regularly takes trips through the same 11 cities. On any given trip, he may stay overnight in a city more than once and he need not stay in every city.

(a) What's the fewest number of nights must he be out to be **sure** that he stays in some city at least twice?

- $\lceil \frac{n}{11} \rceil$ = 2 **Answer = 12**

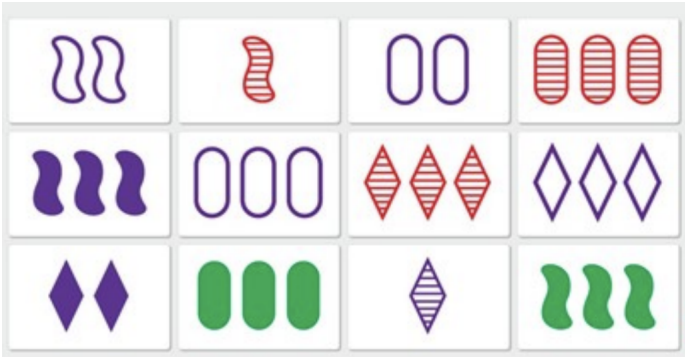
(b) If he takes a trip that has 30 nights out, what the largest number of nights that he knows he must stay in the same city?

- $\lceil \frac{30}{11} \rceil$ = 3 **Answer = 3**

Problem 8. (**32 points**) For this problem, we're going to consider the game SET. If you'd like to try it out yourself, you can play SET free online here: There is a free online version available at `https://www.setgame.com/set/puzzle`.

Each card varies in **four** features, with **three** possibilities for each feature: number (1/2/3), shape (diamond/squiggle/oval), shading (solid/open/striped), and color (red/green/purple).

Each of the 81 possible cards appears exactly once in the deck. Three different cards form a SET if for each of the four features, the three cards are ALL the same or the three cards are ALL different.



For example here, the 3 cards in the rightmost column are a set. They all have 3 shapes on them. However, the three red cards in this graphic would NOT be a set. Try online if you'd like to get the hang of the game before completing these problems.

Not really the answer but context for the answers: It may help to visualize this as a problem with trinary numbers, rather than abstract set cards:

Trinary digit	color	number of shapes	shape	shape filling
0	red	1	diamond	solid
1	green	2	S-shape	open
2	purple	3	oval	striped

In this formulation, 0000 would be a single, red, solid diamond and 1210 would be the solid 3-s shapes in the bottom right hand corner of the graphic above. (1=green, 2=3 shapes, 1=S-shape, 0=solid)

1. How many individual cards share 3 out of 4 attributes with a card that has 2 solid, purple diamonds on it?
   - **Answer:** 8
   - in our ternary example, this is the same as asking "how many ternary numbers share 3 out of 4 digits with the number 2100?
   - in which case we have 2 options where just the leftmost is different, 2 where just the second on the left is different, 2 where just the third from the left is different, and 2 where the rightmost is the only digit different. 2 * 4 = 8
   - Or...just count em
     - **different number**: 1 solid purple diamond, 3 solid purple diamond
     - **different color**: 2 solid red diamond, 2 solid green diamond
     - **different shading**: 2 empty purple diamond, 2 striped purple diamond
     - **different shape**: 2 solid purple oval, 2 solid purple s-shape
2. How many individual cards share nothing in common with a card that has 2 solid, purple diamonds on it?
   - **Answer:** 16

- There are 2 options for each "place" in our ternary number. 0,1, 0, 2, 1,2,1,2. The cardinality of the cartesian product of these 4 sets is $2^4 = 16$

3. Lets say I picked out the card with 2 solid purple diamonds and the card with 2 solid red diamonds. How many sets can be formed with these 2 cards?
   - **Answer:** 1
   - There's only ever 1 card in the deck that is different in only color and it's the 2 solid green diamonds, any other card would violate one of the rules of a set, Either by being too similar to one card (1 solid purple diamond) or too different from both (3 striped green S shapes)
4. Say that I made a deck out of **only** the cards that were solidly filled in and had diamond shapes. How many sets are possible with this mini-deck?
   - *The ternary is no longer useful for this one since we're considering 3 cards rather than 1*
   - First, there would be 9 cards in this mini-deck. (3 cards with 1 solid diamond, 3 cards with 2 solid diamonds, and 3 cards with 3 solid diamonds)
   - There are 3 sets where the count is the same but color differs (1 of all the single diamonds, 1 of all the 2 diamonds, 1 of all the 3 diamonds)
   - There are 3 sets where the count differs but color stays the same (1 of all the red diamonds, 1 of all the green diamonds, 1 of all the purple diamonds)
   - There 6 sets where all 3 are different counts and colors
   - there are
     - 1R, 2G, 3P
     - 1R, 2P, 3G
     - 1G, 2R, 3P
     - 1G, 2P, 3R
     - 1P, 2G, 3R
     - 1P, 2P, 3G
   - **Answer: there are 12 total sets possible**